

OCR Pipeline Architecture

Objective

Convert scanned documents and engineering drawings into structured text.

Pipeline

```
graph TD; A[PDF/Image] --> B[Document Classifier]; B --> C[Docling]; C --> D[PaddleOCR]; D --> E[Layout Detection]; E --> F[Table Extraction]; F --> G[Text Normalization]; G --> H[JSON Output]; H --> I[Metadata Service];
```

Components

Docling

Extracts:

- Text
 - Tables
 - Images
 - Layouts
-

PaddleOCR

Handles

- Scanned PDFs
 - P&ID diagrams
 - Engineering drawings
-

Layout Detection

Identifies:

- Headers
 - Tables
 - Paragraphs
 - Figures
-

Table Extraction

Produces:

```
{
  "table_name": "Maintenance Schedule",
  "rows": []
}
```

Output Schema

```
{
  "text": "",
  "tables": [],
  "images": [],
  "metadata": {}
}
```

Worker

Celery OCR Worker

Document Upload



OCR Queue



OCR Worker



Redis



PostgreSQL